OPERATING SYSTEMS LAB

Experiment 1: Lab Environment Setup

Ubuntu Linux | WSL | VS Code | Python

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Aim: To install, configure and verify Ubuntu Linux (through WSL), Visual Studio Code and Python, and to run basic test programs to confirm that the Operating Systems Lab environment is ready.

1 Install WSL with Ubuntu

Requirements: Windows 10 (version 2004+, build 19041+) or Windows 11, with virtualization enabled in BIOS.

1. Open **PowerShell as Administrator** and run the install command below.
2. Restart the computer when prompted.
3. Ubuntu opens automatically after restart. Create a **UNIX username and password**.

PowerShell (Admin)

```
wsl --install -d Ubuntu
```

Verify WSL

PowerShell

```
wsl --version
wsl --list --verbose
```

Expected: Ubuntu is listed with **VERSION 2** and STATE **Running**. If it shows version 1, upgrade it:

PowerShell

```
wsl --set-version Ubuntu 2
```

Screenshot: Output of wsl --list --verbose

(paste your screenshot here)

Verify Ubuntu

Ubuntu Terminal

```
lsb_release -a
uname -r
cat /etc/os-release
```

Expected: Ubuntu 22.04 or 24.04 LTS, and a kernel string containing **microsoft-standard-WSL2**. Then update the packages:

Ubuntu Terminal

```
sudo apt update && sudo apt upgrade -y
```

2 Install Visual Studio Code

1. Download VS Code from **https://code.visualstudio.com** (Windows installer).
2. During installation tick **Add to PATH** and **Add 'Open with Code' action**.
3. Open VS Code and install extensions (Ctrl+Shift+X): **WSL** (Microsoft), **Python** (Microsoft) and **Pylance**.

Connect VS Code to WSL

Inside the Ubuntu terminal run the commands below. The first run installs the VS Code Server in WSL, and the bottom-left corner of VS Code should show **WSL: Ubuntu**.

Ubuntu Terminal

```
mkdir -p ~/os_lab && cd ~/os_lab
code .
```

Ubuntu Terminal

```
code --version
```

Screenshot: VS Code window showing 'WSL: Ubuntu' in status bar

(paste your screenshot here)

3 Install and Configure Python

Python 3 usually comes preinstalled in Ubuntu. Install the complete toolchain:

Ubuntu Terminal

```
sudo apt install -y python3 python3-pip python3-venv build-essential
```

Verify versions

Ubuntu Terminal

```
python3 --version
pip3 --version
gcc --version
```

Create a virtual environment

Ubuntu Terminal

```
cd ~/os_lab
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install psutil
```

Note: psutil is used in OS experiments to read process and memory information. In VS Code press **Ctrl+Shift+P**, choose **Python: Select Interpreter** and pick **./venv/bin/python**.

Ubuntu Terminal

```
$ python3 --version && pip3 --version && gcc --version | head -1
Python 3.12.3
pip 24.0 from ~/os_lab/venv/lib/python3.12/site-packages/pip (python 3.12)
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
```

4 Test Programs

Test 1: Basic Python script (hello.py)

hello.py

```
import sys, platform

print("Hello, Operating Systems Lab!")
print("Python version :", sys.version.split()[0])
print("OS             :", platform.system(), platform.release())
```

Run

```
python3 hello.py
```

Ubuntu Terminal

```
$ python3 hello.py
Hello, Operating Systems Lab!
Python version : 3.12.3
OS             : Linux 6.18.44-fc-v140
```

Test 2: OS-specific program (os_test.py)

os_test.py

```
import os
import psutil

print("Process ID (PID)  :", os.getpid())
print("Parent PID (PPID) :", os.getppid())
print("Current directory :", os.getcwd())
print("CPU cores         :", os.cpu_count())
print("Total RAM (MB)    :", psutil.virtual_memory().total // (1024 * 1024))

pid = os.fork()
if pid == 0:
    print("Child process, PID:", os.getpid())
else:
    os.wait()
    print("Parent process, child finished")
```

Run

```
python3 os_test.py
```

Observation: os.fork() works only on a real Linux environment (not native Windows Python), so a successful run confirms that WSL Ubuntu is set up correctly.

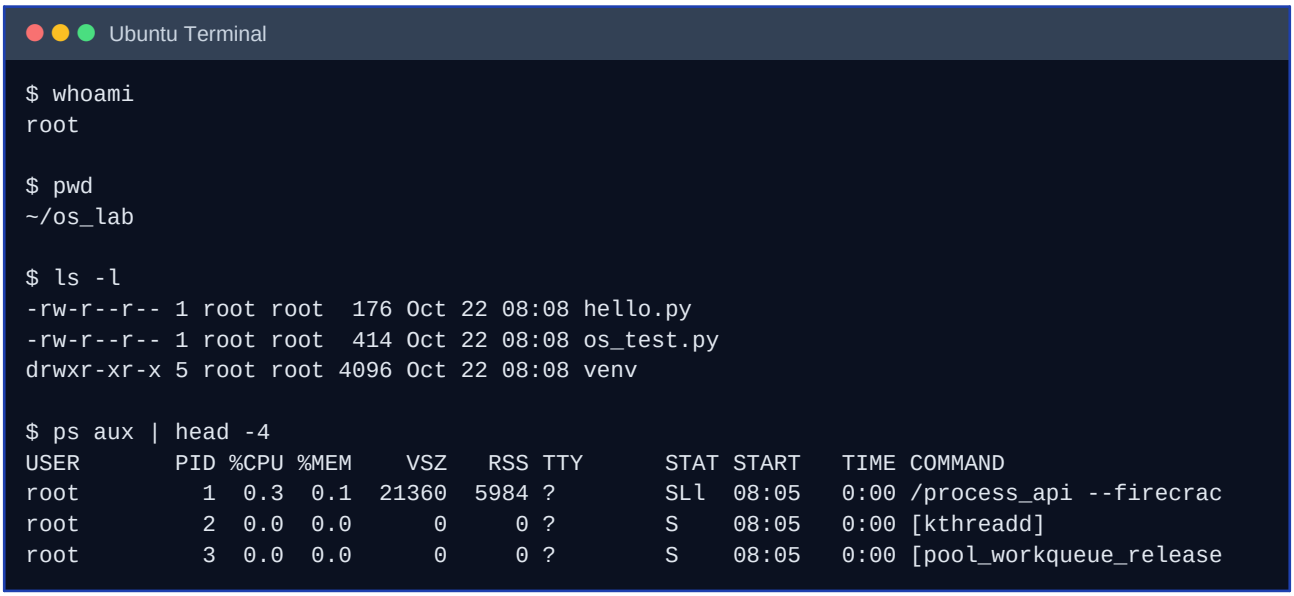
Ubuntu Terminal

```
$ python3 os_test.py
Process ID (PID)  : 198
Parent PID (PPID) : 133
Current directory : ~/os_lab
CPU cores         : 1
Total RAM (MB)    : 3995
Child process, PID: 199
Parent process, child finished
```

Test 3: Basic Linux commands

Ubuntu Terminal

```
whoami
pwd
ls -l
ps aux | head
top -b -n 1 | head -5
```



5 Verification Summary

Component	Command	Expected Result
WSL	`wsl --list --verbose`	Ubuntu, VERSION 2, Running
Ubuntu	`lsb_release -a`	Ubuntu 22.04 / 24.04 LTS
VS Code	`code --version`	Version printed; WSL: Ubuntu in status bar
Python	`python3 --version`	Python 3.10 or above
pip	`pip3 --version`	pip version printed
Test script	`python3 hello.py`	Greeting and versions printed
OS program	`python3 os_test.py`	PID, PPID and fork output

6 Conclusion

Ubuntu was installed through WSL 2, Visual Studio Code was connected to it using the WSL extension, and Python 3 with pip and a virtual environment was configured. All components were verified by checking their versions and by successfully running the test programs. The Operating Systems Lab environment is now ready for further experiments.